

FIFTEEN OPEN PROBLEMS IN GEOMETRIC FLOWS AND GEOMETRIC PDE

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ABSTRACT. We present fifteen open problems in geometric flows and geometric partial differential equations, with an emphasis on higher-order curvature evolution. The problems concern global existence, singularity formation, asymptotic behaviour, regularity and variational rigidity. They include longstanding questions concerning curve diffusion, Willmore flow, biharmonic immersions and the Allen–Cahn equation, together with extensions of recent results in geometric analysis. Each problem is accompanied by a discussion of its motivation, the available results and possible approaches. The collection is motivated by the MATRIX–RIMS Tandem Workshop on geometric flows and related phenomena in September 2026.

INTRODUCTION

Global existence, singularity formation and asymptotic classification are fundamental problems in the analysis of geometric evolution equations. For higher-order curvature flows, the absence of a maximum principle places particular importance on energy estimates, geometric constraints and compactness methods. Closely related questions arise in elliptic rigidity, weak geometric evolution and free-boundary regularity.

The present collection brings together fifteen problems concerning these themes, motivated by the MATRIX–RIMS Tandem Workshop *Geometric flows and related phenomena*, 14–18 September 2026 [1]. The topics range from continuation and stability for curvature flows to critical regularity, selection in weak evolutions, concentration of energy and variational characterisations of distinguished geometric objects. Established conjectures are considered alongside extensions of recent theorems. Each problem is accompanied by relevant literature, a summary of known results and a discussion of the analytic difficulties and possible approaches.

Date: 11 September 2026.

2020 Mathematics Subject Classification. Primary 53E40; Secondary 35K25, 35J60, 35R35, 53C42.

Key words and phrases. Geometric flows, geometric partial differential equations, open problems, curve diffusion, Willmore flow, singularities, regularity.

1. GIGA'S CONJECTURE FOR EMBEDDED CURVE DIFFUSION

Relevant literature.

- G. Wheeler, *Convergence for global curve diffusion flows* [2].
- T. Miura, *Asymptotic circularity of immortal area-preserving curvature flows* [3].
- K.-S. Chou, *A blow-up criterion for the curve shortening flow by surface diffusion* [4].

Problem. Let a smooth closed planar curve evolve by

$$\partial_t \gamma = -k_{ss} \nu$$

on its maximal smooth interval $[0, T)$. If every time slice is embedded, must $T = \infty$?

Motivation. Curve diffusion admits no maximum principle preserving embeddedness. Giga's conjecture asserts that embeddedness throughout the evolution nevertheless excludes finite-time singularities. It would therefore identify loss of embeddedness as a necessary precursor to finite-time breakdown.

Known results. Solutions arising from sufficiently small perturbations of circles exist globally. Miura revisits and extends the global convergence result of [2], correcting gaps in the original arguments and proving smooth exponential convergence of every immortal solution to a round, possibly multiply covered circle; see Theorem 1.1, Remark 2.2 and Section 3.2 of [3]. Giga's conjecture is formulated in [2]. The remaining question is whether embeddedness suffices to extend the flow beyond every finite time.

Possible approach. Chou's continuation criterion implies that, for nonzero enclosed area, finite-time breakdown forces divergence of the L^2 -curvature [4]. A possible approach is to combine area conservation and length dissipation with local curvature estimates and an additional estimate exploiting embeddedness. In the Type I regime, such an argument would require compactness of rescaled solutions and control of possible self-contact in their limits. Type II concentration presents a further obstruction.

2. THE CRITICAL PARAMETER FOR CIRCULAR STABILITY: $c = 3/2$ **Relevant literature.**

- T. Miura and G. Wheeler, *Scale-critical curve diffusion flows* [5].

Problem. For the flow

$$\partial_t \gamma = -(k_{ss} + ck^3)N,$$

with the inward-normal convention, determine the nonlinear stability of the embedded expanding circle at $c = 3/2$, modulo similarities and reparametrisation.

Motivation. The parameter $c = 3/2$ marks a transition in circular relaxation within a family containing curve diffusion, Chen flow and free elastic flow. The dynamics at this parameter may exhibit slower convergence or a bifurcation to noncircular solutions.

Known results. In [5], circular stability was proved for $c < 3/2$. For $c > 3/2$, arbitrarily small perturbations can exhibit increasing curvature

oscillation at the initial time. This initial growth does not determine the long-time dynamics. At $c = 3/2$, the second Fourier harmonic is neutral and nonlinear stability remains unresolved.

Possible approach. The leading nonzero nonlinear term in the amplitude equation for the second harmonic is a natural object of study. Its computation must account for coupling to the fourth harmonic and for scale normalisation. Restriction to reflection-symmetric perturbations may simplify this calculation; a centre-manifold argument would then be required to justify the reduced dynamics.

3. STABILITY OF CONICAL SURFACE-DIFFUSION PINCHOFF UNDER NON-AXISYMMETRIC PERTURBATIONS

Relevant literature.

- G. Wheeler, *Pinchoff by surface diffusion* [6].

Problem. Determine the unstable dimension of the fundamental conical pinchoff profile under general three-dimensional perturbations. Does the same local profile occur for an open set of nearby closed embedded surfaces, after allowing the singular point and time to vary?

Motivation. Stability would establish whether the conical profile describes neck rupture for an open class of initial surfaces. A positive unstable dimension would instead quantify the restrictions on perturbations compatible with this singularity model.

Known results. In [6], closed embedded tori were constructed whose evolutions pinch at one point while remaining embedded before the singular time. The construction establishes a fourth-root waist law and convergence to a conical shrinker through finite-dimensional mode selection. Stability under general perturbations remains unresolved.

Possible approach. The azimuthal Fourier decomposition of the linearised operator separates the non-axisymmetric perturbations into distinct sectors. A spectral analysis would need to distinguish modes generated by geometric symmetries from unstable modes, determine the spectrum in the lowest non-axisymmetric sector and establish coercivity at high angular frequencies. These estimates would provide a basis for a nonlinear stability analysis.

4. CONVERGENCE TO FLAT-CORE p -ELASTICAE

Relevant literature.

- T. Miura and K. Yoshizawa, *Stability of flat-core pinned p -elasticae* [7].

Problem. For planar pinned curves with fixed endpoints and length, consider the constrained $L^2(ds)$ -gradient flow of $\int |k|^p ds$, $p > 2$. Does a solution initially close to a stable flat-core configuration converge to a single stationary curve? Determine the topology and rate of convergence.

Motivation. The evolution is degenerate along zero-curvature segments. Translations of the curved components may also generate a continuum of equilibria, so energy dissipation alone need not exclude drift along the stationary family.

Known results. Miura–Yoshizawa establish the variational stability classification: planar quasi-alternating flat-core configurations are local minimisers, whereas flat-core configurations are unstable in ambient dimension at least three [7]. Convergence of the planar flow near these minimisers remains unresolved.

Possible approach. The case $p = 4$, with one curved component between two straight segments of positive length, provides a restricted setting in which to study the positional modulus. An energy-dissipating evolution and an energy–slope inequality relative to the stationary family could yield convergence, provided that suitable compactness estimates control drift and strengthen subsequential convergence to convergence to a single equilibrium.

5. CRITICAL REGULARITY FOR FORCED PLANAR BRAKKE FLOW

Relevant literature.

- Y. Liu and Y. Tonegawa, *Existence of curvature flow with forcing in a critical Sobolev space* [8].
- K. Tashiro, *A note on generalized BV mean curvature flow with a critical forcing term* [9].

Problem. For planar Brakke flow with velocity $v = h + u^\perp$ and ambient forcing

$$u \in L_t^\infty L_x^2 \cap L_t^2 W_x^{1,2},$$

find a scale-invariant criterion yielding a single graphical representation in a spacetime neighbourhood of a density-one point. Any required local persistence or no-mass-drop hypotheses should be specified. Determine the optimal regularity under such a criterion.

Motivation. This forcing class corresponds to the natural energy bounds for transport by a two-dimensional fluid. At the critical scale, the forcing need not vanish under rescaling, which obstructs regularity arguments based on an unforced limiting flow.

Known results. Liu–Tonegawa establish existence in this forcing class and leave the corresponding regularity theory open [8]. Tashiro proves a BV formula for area change and a compactness result [9]. The theorem announced by Motegi in [1] assumes spatial C^1 graphicality; deriving such a representation from the weak formulation remains a separate regularity problem.

Possible approach. A possible criterion combines small geometric excess, summable Dini-type decay of the local forcing energy and a local no-mass-drop condition. Trace estimates and the BV identity may then support an excess-decay iteration. The principal steps are to establish graphicality and subsequently derive second-order estimates. Quantitative decay estimates for the excess and forcing energy may also yield a Hölder modulus for the tangent.

6. UNIQUENESS OF COLLAPSED ANCIENT RICCI FLOWS OF BOUNDED GIRTH

Relevant literature.

- T. Bourni, T. Buttsworth, R. Lafuente and M. Langford, *Ancient Ricci flows of bounded girth* [10].

Problem. For $n \geq 4$, is the Bourni–Buttsworth–Lafuente–Langford solution the unique $O(n-1)$ -invariant ancient Ricci flow on S^n with positive curvature operator and uniformly bounded girth, up to isometry, parabolic scaling and time translation? Here girth denotes the loop-sweepout width defined in [10].

Motivation. Collapsed ancient solutions lie beyond the scope of noncollapsed classification theorems. Uniqueness would identify a distinguished collapsed model and establish additional symmetry as a consequence of the ancient condition.

Known results. Bourni–Buttsworth–Lafuente–Langford construct an $O(2) \times O(n-1)$ -invariant solution and determine its backward asymptotics [10]. Their Conjecture 1.2 asserts uniqueness under the weaker $O(n-1)$ -symmetry assumption. The analogous statement in dimension three is obstructed by the ancient hypersausage.

Possible approach. Uniqueness within the $O(2) \times O(n-1)$ -invariant class is a natural intermediate problem. A comparison of warped metric profiles in a fixed gauge would require estimates near the ends based on the backward asymptotic models, together with control of the scaling and time-translation freedoms. Extension to the $O(n-1)$ -invariant class would additionally require a symmetry-improvement argument recovering the $O(2)$ action.

7. BOUNDARY REGULARITY FOR MONGE–AMPÈRE EQUATIONS WITH ANISOTROPIC DEGENERACY

Relevant literature.

- Q. Han, J. Liu and Y. Zhou, *Global $C^{1,\alpha}$ regularity for Monge–Ampère equations on planar convex domains* [11].

Problem. Determine sharp boundary-data compatibility conditions and gradient moduli for convex solutions of $\det D^2u = f$, $u|_{\partial\Omega} = \varphi$, when different tangent directions have different finite orders of boundary degeneracy. A model case is a bounded convex domain locally above

$$x_3 = x_1^4 + x_2^6.$$

Motivation. Anisotropic boundary degeneracy introduces competing geometric scales. Sharp regularity estimates would extend the theory beyond uniformly convex domains and characterise the boundary derivatives governing the regularity of the solution.

Known results. Han–Liu–Zhou establish planar global $C^{1,\alpha}$ estimates under finite-order degeneracy and boundary-jet compatibility conditions, with counterexamples demonstrating the necessity of compatibility [11]. Their Remark 1.2 identifies higher-dimensional anisotropic degeneracy as an open extension.

Possible approach. For the homogeneous equation $f = 0$, convex-envelope methods may allow the boundary jets to be classified under the anisotropic dilation $(x_1, x_2, x_3) \mapsto (r^{1/4}x_1, r^{1/6}x_2, rx_3)$. The construction of sharp examples would help identify the required compatibility conditions. An extension to positive densities would require estimates for sections adapted to these anisotropic scales.

8. SINGULAR CONTACT POINTS IN GAUSS-CURVATURE FLOW WITH OBSTACLES

Relevant literature.

- K.-A. Lee and T. Lee, *Gauss curvature flow with shrinking obstacle* [12].

Problem. For convex K^α -flow constrained by a smooth slowly shrinking convex obstacle, classify rescaled limits at contact points where uniform thickness of the coincidence set fails. Determine whether these limits are unique and whether the singular part of the moving contact boundary admits a dimension bound.

Motivation. Changes in the geometry of the contact set may occur at points where a regular free-boundary description fails. Classification of these points is necessary for an analysis of the interaction between the evolving hypersurface and the obstacle.

Known results. Lee–Lee prove global existence and optimal solution regularity for their shrinking-obstacle problem [12]. Their free-boundary C^1 theorem assumes uniform thickness and $\alpha \leq 1/n$, where n is the hypersurface dimension. The structure of contact points at which uniform thickness fails remains unresolved.

Possible approach. Surfaces in $\mathbb{R}^3$, with $0 < \alpha \leq 1/2$, provide a restricted setting near an isolated first-contact point. Freezing the obstacle geometry and deriving the equation for the separation may permit a classification of quadratic rescaled limits. Quantitative convergence and stratification would then require estimates controlling the deviation from these models.

9. SELECTION OF JUMPS IN RATE-INDEPENDENT DROPLET EVOLUTION

Relevant literature.

- W. M. Feldman, I. C. Kim and N. Požár, *An obstacle approach to rate-independent droplet evolution* [13].
- W. M. Feldman, I. C. Kim and N. Požár, *On the geometry of rate-independent droplet evolution* [14].

Problem. In the rate-independent Bernoulli model of droplet motion with contact-angle hysteresis, identify geometric conditions weaker than strong star-shapedness under which minimising movements converge to the obstacle evolution. Assume positive Lipschitz boundary driving with finitely many monotonicity intervals. If approximation limits differ, construct examples and characterise the associated jump-selection mechanisms.

Motivation. Static contact-angle conditions do not determine the depinning times or the discontinuous changes in the wetted region. A selection principle for jumps is therefore an essential component of the evolution problem.

Known results. Feldman–Kim–Požár define a unique obstacle evolution for general data, including non-star-shaped configurations [13, 14]. They prove near-optimal contact-line regularity and convergence of minimising movements in a sufficiently regular, strongly star-shaped setting, with compatible star centres for the source and wetted region. Convergence of the approximation schemes and the associated selection of jumps remain to be understood under weaker geometric hypotheses.

Possible approach. A planar geometry with two competing depinning sites may permit explicit comparison of obstacle evolution, minimising movements and vanishing-viscosity limits. A local comparison principle, valid while the relevant local geometry remains regular, could isolate the geometric mechanism responsible for different jump selections.

10. DEGENERATE MULTI-BUBBLE DYNAMICS FOR THE ENERGY-CRITICAL HEAT EQUATION

Relevant literature.

- K. Kim and F. Merle, *Rigidity results in multi-bubble dynamics for non-radial energy-critical heat equation* [15].

Problem. For global finite-energy solutions of the energy-critical heat equation

$$u_t = \Delta u + |u|^{4/(N-2)}u, \quad N \geq 7,$$

assume the sequential localisation and non-collision hypotheses of Kim–Merle [15] as $t \rightarrow \infty$. Classify the more-degenerate configurations of $J \geq 5$ signed ground-state bubbles not covered by their theorems, and determine whether further universal concentration rates occur.

Motivation. The problem concerns whether the known concentration rates exhaust the possible asymptotic regimes or belong to a larger hierarchy determined by the geometry and signs of bubble interactions. The analysis may also inform the study of concentration in geometric flows.

Known results. Kim–Merle classify radial dynamics and identify three nonradial asymptotic regimes, including new universal rates [15]. Under their hypotheses, these regimes cover all configurations with at most four bubbles and certain configurations with arbitrarily many bubbles. They also exclude finite-time non-colliding bubbling.

Possible approach. For five bubbles, cancellation of the leading interactions defines a finite-dimensional algebraic problem. Expansion of the modulation equations to the first nonvanishing order may produce candidate concentration rates. Their realisation by solutions of the PDE would require a construction compatible with dissipation and the signs of the interactions. Symbolic computation may assist in identifying the possible cancellations.

11. THE CRITICAL CATENOID AND THE FOURTH WIDTH OF THE BALL

Relevant literature.

- D. Ko and G. Shao, *Free boundary minimal annuli in convex balls* [16].

Problem. Prove

$$\omega_4(\mathbb{B}^3) = \text{Area}(\mathbb{K}),$$

where ω_4 is the fourth Almgren–Pitts width and $\mathbb{K}$ is the critical catenoid.

Motivation. This identity would determine the first width beyond those realised by equatorial discs and strengthen the analogy between the critical catenoid in the ball and the Clifford torus in the sphere.

Known results. Ko–Shao construct a four-parameter family whose Simon–Smith width equals the area of the critical catenoid [16]. They obtain

$$\pi < \omega_4(\mathbb{B}^3) \leq \text{Area}(\mathbb{K}).$$

The corresponding Almgren–Pitts lower bound remains open.

Possible approach. An analysis of a hypothetical min–max limit with smaller area would require control of multiplicity, topology and Morse index. Rigidity for the admissible low-index limits, or a deformation excluding non-catenoidal candidates, could yield the required lower bound. Such an argument must apply to arbitrary four-sweepouts.

12. FINITE-TIME SINGULARITIES OF CLASSICAL WILLMORE FLOW

Relevant literature.

- E. Kuwert and R. Schätzle, *Removability of point singularities of Willmore surfaces* [17].
- A. Dall’Acqua, M. Müller, R. Schätzle and A. Spener, *The Willmore flow of tori of revolution* [18].
- E. Mäder-Baumdicker and J. Seidel, *The Willmore energy landscape of spheres and avoidable singularities of the Willmore flow* [19].

Problem. For a smooth immersion $f_0 : \Sigma \rightarrow \mathbb{R}^3$ of a closed, connected, oriented surface, consider the unconstrained L^2 -gradient flow of

$$\mathcal{W}(f) = \frac{1}{4} \int_{\Sigma} H^2 d\mu,$$

where $H = \kappa_1 + \kappa_2$. Does there exist an initial immersion with finite maximal smooth existence time? A restricted version of the question concerns rotationally symmetric immersions of the sphere.

Motivation. The existence of finite-time singularities is a fundamental question for this fourth-order geometric flow. Such singularities would necessitate a theory of continuation beyond finite maximal times. Conversely, global smooth existence for arbitrary initial immersions would require a regularising mechanism independent of a maximum principle.

Known results. The Willmore flow of an immersed sphere with energy below 8π exists globally and converges to a round sphere [17]. Known constructions force singular behaviour without determining whether the singular time is finite or infinite; these include tori with energy arbitrarily close to 8π from above [18, 19]. The existence of finite-time singularities for the unconstrained closed-surface flow remains unresolved.

Possible approach. An axisymmetric catenoidal neck provides a possible model for concentration. A modulation equation for the neck scale, coupled to weighted curvature estimates for the profile curve, may determine whether the scale can vanish in finite time. Control of the outer region is necessary to distinguish local neck concentration from expansion of the entire surface.

13. THE HIGHER-GENUS WILLMORE–KUSNER CONJECTURE

Relevant literature.

- R. Kusner, *Comparison surfaces for the Willmore problem* [20].
- M. Bauer and E. Kuwert, *Existence of minimizing Willmore surfaces of prescribed genus* [21].
- F. C. Marques and A. Neves, *Min-Max theory and the Willmore conjecture* [22].
- R. Kusner, Y. Lü and P. Wang, *The Willmore problem for surfaces with symmetry* [23].

Problem. For each $g \geq 2$, prove that every smooth closed embedded genus- g surface $M \subset S^3$ satisfies

$$W(M) := \int_M (1 + H^2) d\mu \geq \text{Area}(\xi_{g,1}),$$

where $H = (\kappa_1 + \kappa_2)/2$ and $\xi_{g,1}$ is Lawson’s minimal surface. Is equality attained only by Möbius images of $\xi_{g,1}$?

Motivation. The conjecture identifies the minimisers of the Willmore energy for each genus and extends the genus-one Willmore conjecture. It relates a fourth-order Euler–Lagrange equation to the geometry of minimal surfaces and global variational bounds [20].

Known results. Bauer–Kuwert establish the existence of minimising Willmore surfaces of prescribed genus [21], and Marques–Neves prove the genus-one conjecture [22]. Kusner–Lü–Wang establish the asserted minimisation and rigidity under specified Lawson symmetry assumptions [23]. The unrestricted higher-genus statement remains open.

Possible approach. In genus two, the fourth-order second variation may provide a means of analysing symmetry-breaking perturbations and establishing quantitative rigidity near Lawson’s surface. A global argument would additionally require compactness and an energy gap excluding competitors outside a neighbourhood of the candidate modulo Möbius transformations. Weakening the symmetry assumptions in the existing comparison theorem is a related intermediate problem.

14. CHEN’S EUCLIDEAN BIHARMONIC CONJECTURE

Relevant literature.

- S. Montaldo, *An invitation to biharmonic and biconservative submanifolds* [24].
- Y. Fu, M.-C. Hong and X. Zhan, *Biharmonic conjectures on hypersurfaces in a space form* [25].
- K. Akutagawa and S. Maeta, *Biharmonic properly immersed submanifolds in Euclidean spaces* [26].
- Y.-L. Ou and L. Tang, *On the generalized Chen’s conjecture on biharmonic submanifolds* [27].

Problem. Let $f : M^m \rightarrow \mathbb{R}^N$ be a smooth immersion with induced metric $g = f^*g_{\text{eucl}}$. Does

$$\Delta_g^2 f = 0 \quad \implies \quad \Delta_g f = m\mathbf{H} = 0?$$

Here $\mathbf{H}$ is the mean-curvature vector and the Laplacian acts componentwise. No completeness assumption is imposed.

Motivation. The conjecture asserts that every biharmonic Euclidean immersion is minimal. It therefore concerns the rigidity imposed by the induced metric and the normal-bundle geometry on a fourth-order elliptic system.

Known results. Biharmonic hypersurfaces in Euclidean spaces of ambient dimension at most six are known to be minimal [25]. In arbitrary codimension, every properly immersed biharmonic submanifold is minimal [26]. The general case of biharmonic surfaces in $\mathbb{R}^4$ remains unresolved; see [24]. The corresponding assertion for arbitrary nonpositively curved ambient spaces is false [27].

Possible approach. For surfaces in $\mathbb{R}^4$, conformal coordinates allow the tangential and normal components of $\Delta_g \mathbf{H} = 0$ to be coupled to the Gauss–Codazzi–Ricci equations. Estimates controlling variation of the mean-curvature direction may lead to local rigidity. The principal difficulty is to obtain such control without assuming that the mean-curvature vector is parallel.

15. DE GIORGI’S CONJECTURE FOR MONOTONE ALLEN–CAHN SOLUTIONS

Relevant literature.

- N. Ghoussoub and C. Gui, *On a conjecture of De Giorgi and some related problems* [28].
- L. Ambrosio and X. Cabré, *Entire solutions of semilinear elliptic equations in $\mathbb{R}^3$ and a conjecture of De Giorgi* [29].
- O. Savin, *Regularity of flat level sets in phase transitions* [30].
- M. del Pino, M. Kowalczyk and J. Wei, *On De Giorgi’s conjecture in dimension $N \geq 9$* [31].
- E. Florit-Simon and J. Serra, *On stable solutions to the Allen–Cahn equation with bounded energy density in $\mathbb{R}^4$* [32].

Problem. For $4 \leq n \leq 8$, let $u \in C^2(\mathbb{R}^n)$ satisfy

$$\Delta u = u^3 - u, \quad |u| < 1, \quad \partial_{x_n} u > 0.$$

Must $u(x) = \tanh((e \cdot x - a)/\sqrt{2})$ for some unit vector e with $e_n > 0$ and $a \in \mathbb{R}$? Neither pure-phase limits at infinity nor an energy-growth bound are assumed.

Motivation. De Giorgi’s conjecture relates the rigidity of diffuse transition layers to that of minimal graphs. Its dimensional threshold connects elliptic stability, geometric measure theory and singular-interface limits.

Known results. Ghoussoub–Gui and Ambrosio–Cabré resolve dimensions two and three, respectively [28, 29]. Savin proves the conclusion through dimension eight under the additional limits $u(x', t) \rightarrow \pm 1$ as $t \rightarrow \pm\infty$ [30]; del Pino–Kowalczyk–Wei construct counterexamples in dimensions at least nine [31]. Florit-Simon–Serra classify stable solutions in $\mathbb{R}^4$ under cubic energy growth [32]. The monotonicity problem without these additional assumptions remains open in dimensions four through eight.

Possible approach. In dimension four, one possible approach is to derive the energy bound

$$E(R) := \int_{B_R} \left(\frac{1}{2} |\nabla u|^2 + \frac{1}{4} (1 - u^2)^2 \right) dx \leq CR^3 \quad (R \geq 1)$$

from monotonicity. Since monotonicity implies stability, this estimate would place the solution within the classification of [32]. Analysis of the ordered limiting profiles in the monotone direction may identify the conditions needed to control the energy growth.

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